Magma & LMFDB Mini-Course

Hunt 4: Modular Forms

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We will find a non-negative integer n. Each of the four core problems below has a single non-negative integer answer n_i. The bonus is harder and does not count toward the sum. Set

$$n \;=\; n_4 - n_1 - n_2 - n_3.$$

Solutions in Magma or Sage are both welcome.

Problem 4.1 (Smallest non-elliptic level). Let n_1 be the smallest positive integer N for which $S_2^{\text{new}}(\Gamma_0(N))$ contains a newform of dimension ≥ 2 (equivalently, a newform with Hecke field a proper extension of $\mathbb{Q}$).

Problem 4.2 (The Hecke field). At the level n_1 found in Problem 4.1, the relevant newform has Hecke field K_f – a real quadratic field (which would correspond, via Eichler–Shimura, to an abelian surface A_f with real multiplication by an order in K_f). Let $n_2 = \operatorname{disc} K_f$ (a positive integer, since K_f is real).

Problem 4.3 (Smallest icosahedral form). Let N be the smallest positive integer for which $S_1^{\text{new}}(\Gamma_1(N))$ contains a newform f such that the projective image of its associated Galois representation $\rho_f : \operatorname{Gal}(\overline{Q}/Q) \to \operatorname{GL}_2(\mathbb{C})$ is isomorphic to A_5 (the rotational symmetry group of the icosahedron). Let $n_3 = [K_f : \mathbb{Q}]$ be the degree of its Hecke field.

Problem 4.4 (Theta as a modular form – a Lecture 1 callback). The space $M_{12}(\operatorname{SL}_2(\mathbb{Z}))$ of weight-12 modular forms for $\operatorname{SL}_2(\mathbb{Z})$ has dimension 2, spanned by E_4^3 and Δ. The theta series of the Leech lattice, $\theta_\Lambda \in M_{12}(\operatorname{SL}_2(\mathbb{Z}))$, therefore admits a unique expression

$$\theta_\Lambda \;=\; a \cdot E_4^3 \;+\; b \cdot \Delta \qquad (a, b \in \mathbb{Z}).$$

Compute $n_4 = |b|$, the absolute value of the coefficient of Δ.

Bonus 4 (Murmurations of newforms, after Zubrilina 2023). The Zubrilina theorem (*Murmurations*, arXiv:2310.07681, Inventiones 2025): for weight-2 newforms ordered by level N with $N \to \infty$, grouped by root number ± 1, the averaged values $a_p/\sqrt{p}$ as functions of p/N tend to an explicit density. This is the first *proved* murmuration phenomenon (companion to HLOP 2022, which is conjectural).

(a) Pick a level range, e.g. $N \in [1000, 2000]$, and pull from the LMFDB all weight-2 newforms in $S_2^{\text{new}}(\Gamma_0(N))$ in that range, with their a_p values for primes $p \in [2, 1000]$ and their root numbers.

(b) For each prime p in your range, compute the averages

$$\mu_\pm(p) \;=\; \frac{1}{\#\{f : w(f) = \pm 1\}} \sum_{f : w(f) = \pm 1} \frac{a_p(f)}{\sqrt{p}}.$$

(c) Plot $\mu_+(p)$ and $\mu_-(p)$ as functions of p/N for various N values (re-binning if necessary). The Zubrilina murmuration – an oscillation with opposite phase between the two root number bins – should be visible.

(d) (Optional, harder.) Compute Zubrilina's predicted density function (the explicit formula from arXiv:2310.07681, Theorem 1.2 or Corollary 1.3) and overlay it on your plot.

(e) (For the brave.) Generalize to varying weight, after BBLL-D (arXiv:2310.07746).